WORKED SOLUTION

STEP III 1995, Question 8

Pure • Vector geometry

STEP III 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q8

The key idea

When a plane cuts a sphere, the circle it makes has its centre at the foot of the perpendicular from the sphere's centre to the plane. So the sphere's centre must lie on the normal line through the centre of *each* circle, and Pythagoras ties its distance along each normal to the sphere's radius. Everything else in the question is a matter of dotting that one vector equation with well-chosen vectors.

Solution

The normal and the distance

The normal to π through X is the line through $\mathbf{x}$ in the direction $\mathbf{n}$:

$$\mathbf{r} = \mathbf{x} + t\,\mathbf{n}, \qquad t \in \mathbb{R}.$$

It meets π where $(\mathbf{x} + t\mathbf{n}) \cdot \mathbf{n} = p$, that is (since $\mathbf{n} \cdot \mathbf{n} = 1$) where $t = p - \mathbf{x} \cdot \mathbf{n}$. The foot of the perpendicular is therefore $\mathbf{x} + (p - \mathbf{x} \cdot \mathbf{n})\mathbf{n}$, and because $\mathbf{n}$ is a unit vector the perpendicular distance of X from π is

$$|\mathbf{x} \cdot \mathbf{n} - p|.$$

A sphere through both circles

Suppose a sphere with centre $\mathbf{c}$ and radius R contains both circles. Fix i. Every point $\mathbf{y}$ of C_i satisfies $|\mathbf{y} - \mathbf{r}_i| = 1$ and $(\mathbf{y} - \mathbf{r}_i) \cdot \mathbf{n}_i = 0$; as $\mathbf{y}$ runs round C_i, the vector $\mathbf{u} = \mathbf{y} - \mathbf{r}_i$ runs through *every* unit vector perpendicular to $\mathbf{n}_i$.

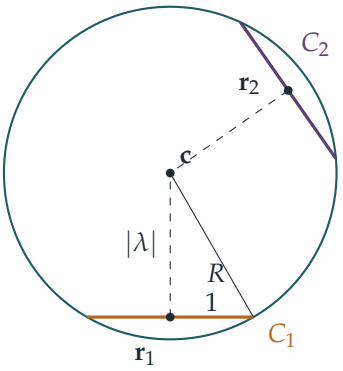
Split $\mathbf{c} - \mathbf{r}_i = \mathbf{a} + \mu_i \mathbf{n}_i$ with $\mathbf{a} \perp \mathbf{n}_i$. Then

$$\begin{aligned} R^2 = |\mathbf{y} - \mathbf{c}|^2 &= |\mathbf{u} - \mathbf{a} - \mu_i \mathbf{n}_i|^2 \\ &= 1 - 2\,\mathbf{u} \cdot \mathbf{a} + |\mathbf{a}|^2 + \mu_i^2, \end{aligned}$$

using $\mathbf{u}\cdot\mathbf{n}_i = \mathbf{a}\cdot\mathbf{n}_i = 0$. The left side is the same for every point of C_i, so $\mathbf{u}\cdot\mathbf{a}$ takes the same value for every allowed $\mathbf{u}$. But $-\mathbf{u}$ is allowed whenever $\mathbf{u}$ is, so $\mathbf{u}\cdot\mathbf{a} = -\mathbf{u}\cdot\mathbf{a} = 0$ for every unit $\mathbf{u}$ in the plane, and choosing $\mathbf{u}$ parallel to $\mathbf{a}$ (if $\mathbf{a} \neq \mathbf{0}$) gives a contradiction. Hence $\mathbf{a} = \mathbf{0}$ and

$$\mathbf{c} = \mathbf{r}_i + \mu_i \mathbf{n}_i, \qquad R^2 = 1 + \mu_i^2.$$

In words: the centre of the sphere lies on the normal to π_i through the centre of C_i, which is exactly the line found in the first part.



Applying this with $i = 1$ and $i = 2$ gives $1 + \mu_1^2 = R^2 = 1 + \mu_2^2$, so $\mu_2 = \pm\mu_1$. Writing $\lambda = \mu_1$,

$$\mathbf{r}_1 + \lambda\mathbf{n}_1 = \mathbf{c} = \mathbf{r}_2 \pm \lambda\mathbf{n}_2,$$

so such a λ must exist whenever a suitable sphere exists, as required.

The two necessary conditions

Rearranging,

$$\mathbf{r}_1 - \mathbf{r}_2 = -\lambda\mathbf{n}_1 \pm \lambda\mathbf{n}_2.$$

Dot with $\mathbf{n}_1 \times \mathbf{n}_2$, which is perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$(\mathbf{r}_1 - \mathbf{r}_2)\cdot(\mathbf{n}_1 \times \mathbf{n}_2) = 0.$$

? **Ask yourself** Where does p_1 come from? It doesn't appear in the vector equation.

The centre $\mathbf{r}_1$ lies in π_1, so $\mathbf{n}_1\cdot\mathbf{r}_1 = p_1$; likewise $\mathbf{n}_2\cdot\mathbf{r}_2 = p_2$. Write $k = \mathbf{n}_1\cdot\mathbf{n}_2$. Dotting $\mathbf{r}_1 + \lambda\mathbf{n}_1 = \mathbf{r}_2 \pm \lambda\mathbf{n}_2$ with $\mathbf{n}_1$ and then with $\mathbf{n}_2$ gives

$$p_1 + \lambda = \mathbf{n}_1\cdot\mathbf{r}_2 \pm \lambda k,$$
$$\mathbf{n}_2\cdot\mathbf{r}_1 + \lambda k = p_2 \pm \lambda.$$

So

$$p_1 - \mathbf{n}_1\cdot\mathbf{r}_2 = \lambda(\pm k - 1), \qquad p_2 - \mathbf{n}_2\cdot\mathbf{r}_1 = \lambda(k \mp 1).$$

With the upper sign both right-hand sides equal $\lambda(k-1)$; with the lower sign they are $-\lambda(k+1)$ and $\lambda(k+1)$. Either way the two left-hand sides are equal or opposite, so

$$(p_1 - \mathbf{n}_1 \cdot \mathbf{r}_2)^2 = (p_2 - \mathbf{n}_2 \cdot \mathbf{r}_1)^2.$$

Geometric meaning

First condition. It says that $\mathbf{r}_1 - \mathbf{r}_2$, $\mathbf{n}_1$ and $\mathbf{n}_2$ are coplanar (the scalar triple product vanishes). Equivalently, the normal to π_1 through the centre of C_1 and the normal to π_2 through the centre of C_2 lie in a common plane: they are not skew, so they meet (at the sphere's centre) or are parallel.

Second condition. By the first part, $|p_1 - \mathbf{n}_1 \cdot \mathbf{r}_2|$ is the distance of the centre of C_2 from π_1, and $|p_2 - \mathbf{n}_2 \cdot \mathbf{r}_1|$ is the distance of the centre of C_1 from π_2. So the condition says these two distances are equal: each circle's centre is as far from the other circle's plane as the other centre is from its plane. This reflects the symmetry of the picture: both circles are at the same distance $|\lambda|$ from $\mathbf{c}$, so a reflection in a suitable plane through $\mathbf{c}$ swaps them, and swaps the two distances.

Checks

- Take $\pi_1 : z = 0$ and $\pi_2 : z = 2$ with unit circles centred on the z-axis. They lie on the sphere centred at $(0, 0, 1)$ of radius $\sqrt{2}$; here $\lambda = 1$ with the lower sign, $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$, and both distances in the second condition equal 2.
- Numerically, we built 5000 random spheres of radius $R > 1$ with two random planes each cutting a unit circle, confirmed every circle point lies on the sphere, and found both conditions held to within 10^{-9} in every case, with both signs occurring. Random unrelated pairs of circles failed the first condition every time.

Common slips

- Assuming the sphere's centre lies on the normal through $\mathbf{r}_i$ without justification; that is the heart of the proof.
- Forgetting the $\pm$: the sphere's centre may be on the same side of both planes as the normals point, or on opposite sides.
- Forgetting that $\mathbf{n}_1 \cdot \mathbf{r}_1 = p_1$ because the centre of C_1 lies in π_1.

Beyond STEP

The conditions here are only necessary: for a full "if and only if" you would also need the two normals actually to meet at a point equidistant from both circles. Questions like this are first steps into analytic geometry of quadrics, where planes, spheres and their intersections are handled with linear algebra; the scalar triple product reappears as a 3×3 determinant measuring volume. Strang's *Introduction to Linear Algebra* develops

these ideas carefully.

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